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MOSER: Modular Open Source Educational Robotics

Expanding Access to Robotics Education to All

Background

Impact of Robotics Education in STEM
Robotics systems are utilized globally to instruct students in critical engineering principles and skills. Systems employed in contemporary classrooms include the REV Robotics Duo System, VEX V5/EXP, Spike Prime, and its predecessor, the EV3. These programs are designed to prepare students for future careers in STEM. Supporting this objective, a meta-analysis on the impacts of robotics in STEM education concluded that "students benefited from robot-assisted STEM education, which mainly promoted their learning performances and attitudes" (Ouyang & Xu, 2024, p. 13). While these systems demonstrably enhance engagement with STEM curricula, their planned obsolescence and proprietary nature pose a significant barrier to local educational institutions.

The Economic Barrier of Implementing Robotics in Schools
The implementation and ongoing maintenance of these robotics systems represent a substantial financial burden for schools. Should a vendor discontinue support for a system, institutions frequently incur losses due to the inability to source appropriate alternatives for necessary replacement or maintenance components. This challenge is exemplified by schools such as Mayfield High School, which adopted the VEX Cortex System. Following the company's decision to cease support and remove critical specifications from its website, the school encountered considerable difficulty in identifying comparable motors and batteries.

The Maintenance Barrier of Discontinued Equipment still Utilized in Schools
VEX products such as the VEX Cortex and previous iterations such as the PIC Microcontroller, regularly have their product pages removed with no official archive available, this drastically reduces the ability for students in schools continuing to utilize the VEX Cortex ecosystem to easily design compatible parts and limiting students from practically applying CAD and related skills. Furthermore, as seen in Table 1 regarding the VEX V5 system, there is no ability to utilize non-VEX products, resulting in a monotonous ecosystem, as such when the VEX V5 system is discontinued. It will become extremely difficult for schools to maintain their systems as parts break due to natural wear and tear, resulting in e-waste as schools transition their equipment. Furthermore, the LEGO Education kits, which are the EV3 and Spike Prime as listed in Table 1, operate on a 6-year lifecycle as demonstrated by the year release of the EV3 and Spike Prime, and the now obsolete NXT, which was released in 2006. This 6-year lifecycle means schools who rely on the LEGO Education continuum are forced to continually invest in new systems every 6 years, else they'll be placed in a position where they're unable to purchase replacement parts like with VEX.

Design Statement & Principles

Design Statement:
Develop a robotics system for application in an educational environment at a significantly lower cost to mass-manufactured comparable robotics system currently in-use. This system should be based on the principles of repairability, accessibility, and longevity.

Guiding Principles:

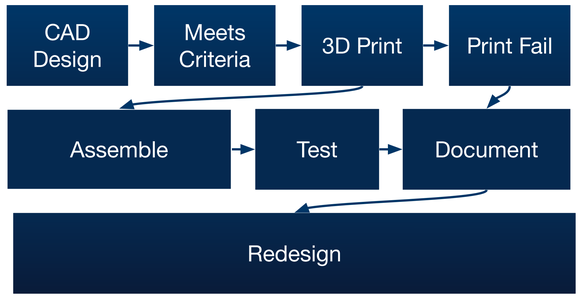
1. The system should be designed to promote the FDM manufacturing of structural parts within the school, limiting costs and enabling greatever creativity within a schools implementation of robotics education
2. Emphasize the usage of readily available electronics and components rather than developing proprietary components and making a single point of failure in the implementation and maintenance of this system
3. Should be open-source, allowing students and educators to develop new additions to the design system, allowing the implementation of a wider range of sensors, control systems, and batteries under a uniform design system.

Design Criteria

Table 2
MOSER Design System Criteria

Name	Criteria
Maker Control System	Utilize a readily available Maker board with specifications and CAD
Sensor Ports	Maker board should support 2 or more sensors utilizing a standardized cable, such as Grove
Servos	Maker board should support at least 4 servos
3D Printed Structure	Large structural pieces should be primarily 3D Printed
Metric Based	The structure should be designed with M4 screws in mind, and easy to CAD using millimeters
Motor Limit	Must support at least 4 motors
Motor Cost	The cost per motor should be under 20$
Programming Language	The programming language of the system should enable students to learn text-based coding and provide low-level access to the maker board

Engineering Methodology



Flowchart was created by Riley Bautista using Google Sheets, 2026

Competitive Analysis

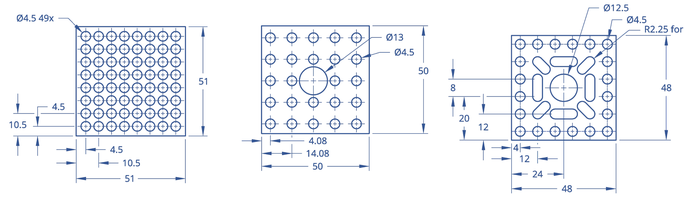
Analysis of current educational robotics systems reveals several critical issues:

1. **System Discontinuation:** All evaluated systems are either discontinued or will be within two years.
2. **Lack of Support after Discontinuation:** Companies offer limited to no replacement parts after discontinuation. VEX Robotics, specifically, removes specifications and engineering drawings, preventing educators from purchasing comparable alternatives.
3. **Proprietary Nature:** All evaluated systems are proprietary and closed-source. Companies are the sole providers of software. If companies restrict access or drop support for programming tools (e.g., when operating systems are updated), schools lose the ability to program the robots.

MOSER addresses these problems by offering a unique system where schools can implement readily available, open-source components. This allows schools to use common programming languages and enables students to develop robots at home due to the broad availability of electronics and access to the MOSER Design System.

As such MOSER is a competitive system for local schools and challenges companies to develop more repairable systems with a focus on long-term utilization within classrooms, rather than a replaceable product, that only schools with large budgets can afford to cycle

Design Iterations



Engineer Drawings were created by Riley Bautista using OnShape, 2026

Within the first iteration, there was an attempt to utilize a grid of M4 holes, however recognizing that since there's no hole for a driving shaft such as the one utilized on the TT Motors, there was less versatility in mounting.

As such the design pivoted to a central hole with space for a 6mm bearing, however I recognized that despite this working, there was still some minor limitations such as being unable to mount it from a 45 degree angle

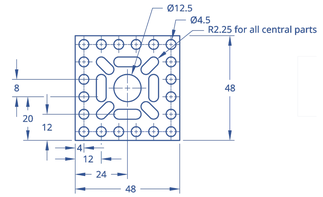
For the final iteration, I developed a pattern based upon M4 screws with the ability to mount at a 45 degree angle

Final Design & System Outline

MOSER in a Nutshell

The MOSER Design System moves away from proprietary plastic molding and expensive aluminum extrusions in favor of 3D Printed design.

1. **Metric-Standard Framework:** The system utilizes a standardized grid based on M4 hardware. This ensures that structural components are not only 3D-printable (FDM) but can also be reinforced with off-the-shelf components found at online retailers such as DigiKey or McMaster-Carr
2. **6mm D-Shaft Standard:** By adopting a 6mm D-shaft for all motion components, MOSER ensures compatibility with a vast ecosystem of industrial and hobbyist gears, pulleys, and wheels, bypassing the "vendor lock-in" seen in VEX or LEGO systems.
3. **FDM-Optimized Geometry:** Parts are designed with print-orientation in mind to maximize tensile strength along stress axes, ensuring that 3D-printed PETG components can withstand the rigors of classroom competition.



MOSER Design Pattern

- Simple
- Adaptable
- Versatile
- Metric-based

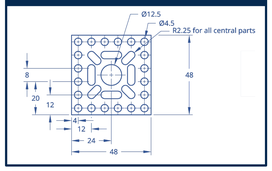
The MOSER Pattern empowers utilization of TT Motors, 6mm bearing, and D-Shafts

Conclusion
The MOSER Design System represents a paradigm shift in educational robotics by replacing the "disposable" nature of proprietary kits with a framework built for longevity. By standardizing on M4 metric hardware and 6mm D-shafts, MOSER decouples a school's curriculum from the restrictive life cycles of corporate vendors like VEX or LEGO. Instead of facing "end-of-life" software lockouts or skyrocketing costs for replacement parts, educators and students are empowered to manufacture their own structural components via FDM 3D printing. This open-source approach not only reduces the financial barrier to entry for schools across the world but also forces a deeper engagement with authentic engineering, teaching students to navigate the complexities of CAD design, digital fabrication, and universal programming languages.

Ultimately, MOSER transforms robotics from a consumer product into a community-owned utility, ensuring that a student's ability to innovate is limited only by their creativity, not by their school's budget.

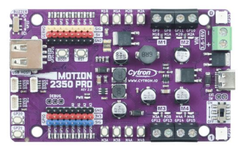
After evaluation of maker components and maker boards. The electronics listed below are the most cost-effective and criteria meeting components available on DigiKey and designed to be utilized in the MOSER System, due to the nature of MOSER, **schools can opt to replace any of these components with another electronic meeting the criteria**

Design Pattern



Engineer Drawing was created by RIley Bautista using OnShape, 2026.

Control System



Graphic obtained from the Internet
From Cytron [Photograph], by Cytron Electronics, 2025 (https://static.cytron.io/image/cache/catalog/products/MOTION-2350-PRO/motion-2350-pro-r-800x800.png)

Battery



Graphic obtained from the Internet
From ServoCity [Photograph], by Base10 Assets, n.d. (https://cdn11.bigcommerce.com/s-f6vfspkkjf/images/stencil/original/products/2158/8184/3100-0006-0001__63888__64068.1701975406.jpg)

Motors



Graphic obtained from the Internet
From DigiKey [Photograph], by Adafruit Electronics, n.d. (https://mm.digikey.com/Volume0/opasdata/d220001/medias/images/876/3777.JPG)

Cost Comparison Table

Table 1
Cost Analysis Between Pre-Existing Comparable Educational Robotics Systems

Name	Cost	Audience	Released yr.	Main Limitation
VEX Cortex	850$	9-12th Grade	2010	Discontinued, batteries no-longer sold, hard to find alternatives, specs previously available on the VEX website are removed.
VEX V5	850$	9-12th Grade	2018	Closed-down firmware, unable to use non-VEX products with the V5
LEGO EV3	675$	6-12th Grade	2013	Discontinued; low processing power; limited to LEGO ecosystem.
LEGO Spike Prime	430$	6-8th Grade	2020	Unable to use non-LEGO motors, end of life in 2026.
REV Robotics Duo	695$	7-12th Grade	2017	High entry barrier, proprietary 'closed' sensor and motor cable standards, likely to be discontinued in 2027

Impact

- MOSER enables cheaper and broader access to robotics education across the world
- Challenges the Educational Robotics Industry through an emphasis on access to replacement parts
- Places pressure on the Educational Robotics industry to create longer-lasting products and provide longer-lasting support
- Enables students to utilize Maker programming tools, allowing them to learn and apply skills learned outside of school without relying on a proprietary and expensive system
- Enable student interaction with industrial utilized parts rather than abstracted and simplified systems
- Enable schools to utilize the MOSER Design System for longer periods of time due to continued maintenance of new motors and control systems
- Enable students and educators to design and develop new parts to be added to MOSER enabling greater creativity
- Enable students to contribute to an open-source project and understand the importance and value of open-source projects/initiatives

References

Table 1 & 2 were created by Riley Bautista in Google Sheets, 2026

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